

5. Eigenvalues and Eigenvectors

5.1 Eigenvectors and Eigenvalues

Definition:

An eigenvector of an $n \times n$ matrix A is a nonzero vector $\vec{x}$ such that $A\vec{x} = \lambda\vec{x}$ for some scalar λ . A scalar λ is called an eigenvalue of A if there is a nonzero solution $\vec{x}$ of $A\vec{x} = \lambda\vec{x}$; such an $\vec{x}$ is called an eigenvector corresponding to λ .

Theorem 1:

The eigenvalues of a triangular matrix are the entries on its main diagonal.

Theorem 2:

If $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p$ are eigenvectors that correspond to distinct eigenvalues of an $n \times n$ matrix A , then the set $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p\}$ is linearly independent.

5.2 The Characteristic Equation

Theorem (The Invertible Matrix Theorem, continued):

Let A be an $n \times n$ matrix. Then A is invertible if and only if:

- s. The number zero is not an eigenvalue of A .
- t. The determinant of A is not zero.

Theorem 3: Properties of Determinants

Let A and B be $n \times n$ matrices.

- a. A is invertible if and only if $\det A \neq 0$
- b. $\det(AB) = (\det A)(\det B)$
- c. $\det(A^T) = \det A$
- d. If A is triangular, then $\det A$ is the product of the entries on the main diagonal of A .
- e.
 - Row replacement operations on A do not change the determinant.
 - A row interchange changes the sign of the determinant.
 - A row scaling also scales the determinant by the same scalar factor.

Characteristic Equation:

A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if it satisfies the characteristic equation:
 $\det(A - \lambda I) = 0$

Theorem 4:

If $n \times n$ matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

Definitions:

- If A is an $n \times n$ matrix, then $\det(A - \lambda I)$ is a polynomial of degree n called the characteristic polynomial of A .
- The (algebraic) multiplicity of an eigenvalue λ is its multiplicity as a root of the characteristic equation.

Definition: Similarity

- Two $n \times n$ matrices A and B are similar if there exists an invertible matrix P such that $P^{-1}AP = B$, or equivalently, $A = PBP^{-1}$.
- Changing A to $P^{-1}AP$ is called a similarity transformation.

5.3 Diagonalization

Definition: Diagonalizable

A square matrix A is **diagonalizable** if A is similar to a diagonal matrix.

Theorem 5 (The Diagonalization Theorem):

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A .

In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .

proof idea

because $AP = PD$, and P is invertible, so $A = PDP^{-1}$

Corollary:

A is diagonalizable if and only if there are enough eigenvectors to form a basis of $\mathbb{R}^n$. Such a basis is an eigenvector basis of $\mathbb{R}^n$.

Theorem 6:

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Theorem 7:

Let A be an $n \times n$ matrix whose distinct eigenvalues are $\lambda_1 \dots \lambda_p$:

- For $1 \leq k \leq p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of the eigenvalue λ_k .
- The matrix A is diagonalizable if and only if:
 - the characteristic polynomial factors completely into linear factors;
 - the dimension of the eigenspace for each eigenvalue λ_k equals the multiplicity of λ_k .
- If A is diagonalizable and B_k is a basis for the eigenspace corresponding to λ_k for each k , then the total collection of vectors in the sets $B_1, B_2, \dots, B_p$ is an eigenvector basis for $\mathbb{R}^n$.

5.4 Eigenvectors and Linear Transformation

Setup

- Let V be an n -dimensional vector space, let W be an m -dimensional vector space, and let T be any linear transformation from V to W .
- We choose bases B (for V) and C (for W) respectively.

Coordinate Vectors & Matrix Representation of T

1. For any $\vec{x} \in V$:

- The coordinate vector $[\vec{x}]_B \in \mathbb{R}^n$;
- The coordinate vector of its image $[T(\vec{x})]_C \in \mathbb{R}^m$.

2. Let $\{\vec{b}_1, \dots, \vec{b}_n\}$ be the basis B for V :

- If $\vec{x} = r_1 \vec{b}_1 + \dots + r_n \vec{b}_n$, then $[\vec{x}]_B = \begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix}$.
- By linearity of T : $T(\vec{x}) = r_1 T(\vec{b}_1) + \dots + r_n T(\vec{b}_n)$.
- By linearity of the coordinate mapping from W to $\mathbb{R}^m$:
 $[T(\vec{x})]_C = r_1 [T(\vec{b}_1)]_C + \dots + r_n [T(\vec{b}_n)]_C$

3. Matrix relation:

- $[T(\vec{x})]_C = M[\vec{x}]_B$, where the matrix
 $M = \begin{bmatrix} [T(\vec{b}_1)]_C & [T(\vec{b}_2)]_C & \dots & [T(\vec{b}_n)]_C \end{bmatrix}$
- M is a matrix representation of T , called **the matrix for T relative to the bases B and C** .

Special Case: B-matrix for T

If $W = V$ and the basis $C = B$:

- The matrix M is called **the matrix for T relative to B** (or simply the B -matrix for T), denoted by $[T]_B = M$.

Theorem 8: Diagonal Matrix Representation

- Suppose $A = PDP^{-1}$, where D is an $n \times n$ diagonal matrix.
- If B is the basis for $\mathbb{R}^n$ formed from the columns of P , then D is the B -matrix for the transformation $\vec{x} \mapsto A\vec{x}$.

Auxiliary Transformation Relation & Note

1. Transformation diagram:

$$\begin{array}{ccc} \vec{x} & \xrightarrow{A} & T(\vec{x}) \\ \downarrow P^{-1} & & \uparrow P \\ [\vec{x}]_B & \xrightarrow{B} & [T(\vec{x})]_B \end{array}$$

2. Derivation:

- From $A = PBP^{-1}$, we get $B = P^{-1}AP$.

3. (note.) Method to find $B = P^{-1}AP$:

- Perform row operations on the block matrix $\begin{bmatrix} P & AP \end{bmatrix}$ to transform it into $\begin{bmatrix} I & P^{-1}AP \end{bmatrix}$.

5.5 Complex Eigenvalues

Definition:

A complex scalar λ satisfies $\det(A - \lambda I) = 0$ if and only if there is a nonzero vector $\vec{x}$ in $\mathbb{C}^n$ such that $A\vec{x} = \lambda\vec{x}$. We call λ a complex eigenvalue and $\vec{x}$ a complex eigenvector corresponding to λ .

Complex Conjugate Vector:

The complex conjugate of a vector $\vec{x}$ in $\mathbb{C}^n$ is the vector $\bar{\vec{x}}$ whose entries are the complex conjugates of the entries in $\vec{x}$.

Real and Imaginary Parts of Complex Vector:

The real and imaginary parts of a complex vector $\vec{x}$ are the vectors $\text{Re}\vec{x}$ and $\text{Im}\vec{x}$ in $\mathbb{R}^n$ formed from the real and imaginary parts of the entries of $\vec{x}$.

Property of conjugates for complex matrix algebra

- $r\bar{\vec{x}} = \bar{r}\vec{x}$
- $\bar{B}\vec{x} = \overline{B\vec{x}}$
- $\bar{BC} = \bar{B}\bar{C}$
- $r\bar{B} = \bar{r}\bar{B}$

Property of Complex Eigenvalues (Real Matrix):

When A is real, its complex eigenvalues and eigenvectors occur in conjugate pairs (i.e., if $\lambda = a + bi$ is an eigenvalue, then $\bar{\lambda} = a - bi$ is also an eigenvalue).

Theorem 9:

Let A be a real 2×2 matrix with a complex eigenvalue $\lambda = a - bi$ ($b \neq 0$) and an associated eigenvector $\vec{v}$ in $\mathbb{C}^2$

$$A = PCP^{-1}, \text{ where } P = [\text{Re}\vec{v} \text{ Im}\vec{v}], C = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

if $r = |\lambda| = \sqrt{a^2 + b^2}$

so

$$C = \begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix} \begin{pmatrix} a/r & -b/r \\ b/r & a/r \end{pmatrix} = \begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix} \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \text{ 表示旋转!}$$

5.6 Discrete Dynamical Systems

Setup:

Suppose A is diagonalizable, with n linearly independent eigenvectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ and corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Suppose the eigenvectors are arranged so that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$.

Solution to Recurrence Relation:

Since $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ forms a basis for $\mathbb{R}^n$, any initial vector $\vec{x}_0$ can be written uniquely as $\vec{x}_0 = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n$. Then:
 $\vec{x}_1 = A\vec{x}_0 = c_1 \lambda_1 \vec{v}_1 + \dots + c_n \lambda_n \vec{v}_n$

...

$$\vec{x}_k = c_1 (\lambda_1)^k \vec{v}_1 + \dots + c_n (\lambda_n)^k \vec{v}_n$$

Graphical Description of Solutions (2D Case):

The graph of $\vec{x}_0, \vec{x}_1, \dots$ is a trajectory of the dynamical system.

$$\text{Suppose } \vec{x}_k = c_1 (\lambda_1)^k \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 (\lambda_2)^k \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

1. $|\lambda_1| < 1$ and $|\lambda_2| < 1$: origin is an attractor.
2. $|\lambda_1| > 1$ and $|\lambda_2| > 1$: origin is a repeller.
3. $|\lambda_1| > 1$ and $|\lambda_2| < 1$: origin is a saddle point.

Change of Variable:

The eigenvectors of A form a basis $\{\vec{v}_1, \dots, \vec{v}_n\}$ for $\mathbb{R}^n$. Let $P = (\vec{v}_1 \dots \vec{v}_n)$ and let D be the diagonal matrix with the corresponding eigenvalues on the diagonal. Given a sequence $\{\vec{x}_k\}$ satisfying $\vec{x}_{k+1} = A\vec{x}_k$, define a new sequence $\{\vec{y}_k\}$ by:

$$\vec{y}_k = P^{-1} \vec{x}_k \quad \text{or} \quad \vec{x}_k = P \vec{y}_k$$

Using $A = PDP^{-1}$, we find that:

$$P \vec{y}_{k+1} = A \vec{x}_k = AP \vec{y}_k = PDP^{-1} P \vec{y}_k = PD \vec{y}_k$$

Dividing both sides by P (since P is invertible):

$$\vec{y}_{k+1} = D \vec{y}_k$$

Power of Matrix A :

$$A^k = PD^k P^{-1}$$